FOCEX agent

User Manual

Force constants and phonons from a crystal structure: VASP · SC-SNAPS · FOCEX · Diagnosis & repair

Keivan Esfarjani

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https://github.com/KeivanS/Anharmonic-lattice-dynamics (FOCEX/AGENT)

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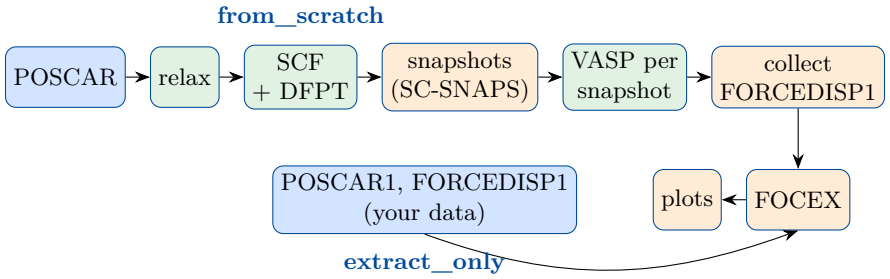
1 What the FOCEX agent does

FOCEX obtains the force constants of a crystal (harmonic, cubic, quartic, ...) by fitting them to the forces that density-functional theory (DFT) gives on the atoms of a supercell in which every atom has been displaced a little at random (a *snapshot*). From the force constants it computes the phonon dispersion, density of states, group velocities, Grüneisen parameters, elastic constants, sound speeds and the thermal expansion; THERMACOND then computes the lattice thermal conductivity from the same files.

Making those forces takes many steps: relax the crystal, orient the cell the way FOCEX expects, compute the Born charges of an insulator, choose a supercell, generate the snapshots, run one DFT calculation per snapshot, collect the forces, write the five input files of FOCEX and run it. Each step has its own input files and its own ways to fail. The FOCEX agent, `focex-agent.py`, does them all from one short *instruction file*, checks the result of every step, writes what it did to `report.log`, and can be restarted at any time: finished steps are not repeated. When a run stops, `diagnose.py` finds out where and why, and writes a script that repairs it (Section 10).

It works in one of two modes:

Mode	You give	The agent does
from_scratch	a crystal structure (`POSCAR`) and the instruction file	VASP relaxation, SCF and Born charges; SC-SNAPS snapshots; one VASP run per snapshot; FOCEX; plots (Section 4)
extract_only	existing supercells and their forces (`POSCAR1`, `FORCEDISP1`, ...) and the instruction file	rotates the data into the FOCEX frame; FOCEX; plots (Section 5)



green: VASP; orange: SC-SNAPS, FOCEX and their tools; blue: your input

2 What you need, and how to install it

You do not need to have used VASP-Flow before: the agent uses it as a library, to write the VASP input files and the batch scripts. This section has everything you need to set up.

2.1 The programs

What	Needed for	How to get it
VASP 5.4 or newer	every DFT step (from_scratch only)	your licence; on a cluster usually a module
POTCAR library	VASP: the PAW potentials, one folder per element	comes with VASP
MPI	running VASP in parallel (`mpirun` or `srun`)	system or cluster module
Python $\geq$ 3.8 with numpy, spglib, pymatgen	the agent and `diagnose.py`	a virtual environment, below
VASP-Flow	writes the VASP inputs and batch scripts	`git clone`, below
FOCEX, `read_outcar.x`	the fit; reading the forces from the OUTCARs	this repository, `make`
SC-SNAPS (`sc_snaps.x`)	the snapshots (from_scratch only)	its own repository, `make`
gnuplot	the plots (PDF)	`brew install gnuplot`, `apt install gnuplot`, or a module
a Fortran compiler	building FOCEX and SC-SNAPS	`gfortran` (GCC)

2.2 Install, once

The same commands on a workstation and on the login node of a cluster (there, load Python 3.8+ and GCC first, e.g. `module load gcc python`).

```
cd ~
# 1. this repository (FOCEX, its utilities, the agent) and SC-SNAPS
git clone https://github.com/KeivanS/Anharmonic-lattice-dynamics.git
git clone https://github.com/KeivanS/SC_SNAPS.git
mkdir -p ~/BIN ~/ALADYN/bin
# FOCEX -> ~/ALADYN/bin/focex-8.19.x (3 min), its utilities (read_outcar.x, ...)
    -> ~/ALADYN/bin
# (the top-level make builds THERMACOND and SCOP8 as well; they need MPI)
(cd Anharmonic-lattice-dynamics/FOCEX && make)
(cd Anharmonic-lattice-dynamics/FOCEX/utility && make)
(cd SC_SNAPS && make)                                    # -> ~/BIN/sc_snaps.x

# 2. VASP-Flow; the agent imports it from ~/VASP-Flow
#    (another place: export VASP_FLOW_DIR=/that/place)
git clone https://github.com/KeivanS/VASP-Flow.git

# 3. one Python environment for VASP-Flow and the FOCEX agent
python3 -m venv ~/dftphonon-env
source ~/dftphonon-env/bin/activate
pip install numpy spglib pymatgen matplotlib

# 4. optional: the commands focex-agent and focex-diagnose (symlinks in
    ~/ALADYN/bin)
(cd Anharmonic-lattice-dynamics && make python-tools)
export PATH=~/ALADYN/bin:$PATH     # put this line in ~/.bashrc
```

The agent finds the programs at these places; a program installed elsewhere is named in the instruction file (`FOCEX_X`, `SC_SNAPS_X`, `READ_OUTCAR_X`, Section 6). With step

4 and the environment active, `focex-agent` and `focex-diagnose` are short for `python .../FOCEX/AGENT/focex-agent.py` and `python .../FOCEX/AGENT/diagnose.py`.

2.3 Tell VASP-Flow where VASP, MPI and the POTCARs are

The POTCAR library is always given by an environment variable. Put it in your `~/.bashrc` (or `~/.zshrc`) so that every job sees it:

```
export VASP_POTCAR_DIR=/path/to/potpaw_PBE.54      # holds Zn/, O/, Ga/, ...
```

VASP and MPI come from a *profile*, a small JSON file in `~/VASP-Flow/profiles/`, chosen with `--profile` (or `PROFILE:` in the instruction file):

profile	use it for
`(none)`	a workstation. VASP from the variables `VASP_STD` (default `~/BIN/vasp_std`) and `MPI_LAUNCH` (default `mpirun -np`), set in `~/.bashrc`; the number of ranks is `MPI:` in the instruction file; `run_workflow.sh` is started with `bash`
`workstation`	the same, with the paths written in `profiles/workstation.json` instead
`slurm`	a template for a SLURM cluster: edit `profiles/slurm.json` (VASP command, `modules` to load, partition, nodes, tasks per node, time, account)
`rivanna`	the University of Virginia cluster Rivanna
your own	copy `slurm.json` to `mycluster.json`, edit it, and use `--profile mycluster`

With a SLURM profile, `run_workflow.sh` begins with the `##SBATCH` lines and the `module load` lines of the profile and is submitted with `sbatch`.

2.4 Check the set-up

```
source ~/dftphonon-env/bin/activate
python ~/VASP-Flow/check_setup.py local      # a workstation;  "slurm" on a cluster
```

It checks Python and its packages, the POTCAR library, VASP and MPI (and on a cluster the profile), and prints for every problem the command that fixes it. **Ready.** at the end means you can start.

2.5 When is the Python environment needed?

It is the same environment as in VASP-Flow's manual: one, `~/dftphonon-env`, serves both (DFT with VASP-Flow, phonons with the FOCEX agent). It is needed whenever *you* start `focex-agent.py` or `diagnose.py`: they import numpy, spglib, pymatgen and VASP-Flow. Either activate it first (`source ~/dftphonon-env/bin/activate`, then `python ...`) or call its interpreter directly (`~/dftphonon-env/bin/python ...`). The scripts the agent writes (`run_workflow.sh`, `resubmit_failed.sh`, `diagnosis/fix_it.sh`) contain the full path of the interpreter you used, so the batch jobs need no activation. The plots need only gnuplot; without it the agent writes the plot scripts and you run `makeplots.sh` later, where gnuplot is.

3 Quick start

The steps of a from_scratch calculation (Section 5 has those of extract_only):

1. **Make a folder** for the material with its crystal structure, in VASP's POSCAR format, named `POSCAR`. Any cell of the crystal will do – primitive or not, any orientation: the agent reduces and orients it.

2. **Copy the instruction file** `FOCEX/AGENT/focex-agent-instructions.txt` into that folder and edit it. Only two lines are required: `Project:` (a name) and `RANKS:` (which force constants to fit, e.g. `1 1 1 0 0 0 0 0` for harmonic and cubic). Set also `FUNCTIONAL:` and `MPI:` (the number of MPI ranks for VASP); every other line is optional and shows its default.

3. **Set up the project**: the agent makes the project folder (`PROJECT_DIR`, by default named after `Project:`) with the relaxation inputs and `run_workflow.sh`. This takes seconds and runs no VASP.

4. **Start the workflow** from the project folder, and follow `report.log`.

5. **If it stops**, run `diagnose.py` (Section 10).

```
cd ~/work/ZnO                        # holds POSCAR
A=~/Anharmonic-lattice-dynamics/FOCEX/AGENT
cp $A/focex-agent-instructions.txt .
nano focex-agent-instructions.txt  # Project, RANKS, FUNCTIONAL, MPI

source ~/dftphonon-env/bin/activate
python $A/focex-agent.py --profile rivanna
cd ZnO
sbatch run_workflow.sh               # without a profile: nohup bash run_workflow.sh
    &
tail -f report.log
```

The options of `focex-agent.py` are all optional:

option	meaning
`--instr FILE`	the instruction file, if it is not called `focex-agent-instructions.txt`
`--coord FILE`	the crystal structure, if it is not called `POSCAR` (from_scratch)
`--profile NAME`	the VASP-Flow profile (previous section); default: `PROFILE:` of the instruction file, else none
`--force`	set up again a project folder that exists already. Its record of finished tasks is reset, so use it only to start over, after removing the old run folders

Note. Threads. You do not have to do anything about OpenMP threads: the agent runs VASP with one thread per MPI rank (`OMP_NUM_THREADS=1`). Without that, a BLAS library built with OpenMP starts one thread per core in every MPI rank and the cores are oversubscribed many times (a 96-atom ZnO supercell made no SCF step in 30 minutes on 8 ranks). Only if your VASP is built for hybrid MPI + OpenMP runs and you want them, set `VASP_OMP_NUM_THREADS` to the number of threads per rank before you start the workflow.

4 Mode from_scratch: from a POSCAR to phonons

`run_workflow.sh` runs `focex-agent.py run <PROJECT_DIR>`, which does the ten tasks below in order. Each finished task is recorded in `state.json` and in `report.log`, with what was checked; a task that fails stops the workflow, and starting `run_workflow.sh` again continues from that task. A job that hits its time limit is simply submitted again.

`relax` The structure you gave is reduced to its primitive cell (Niggli reduction) and rotated into the orientation FOCEX and SC-SNAPS use (Section 4.2); the result is `POSCAR.init`. VASP relaxes the atoms and the cell shape and volume in `01_relax`. If the forces are not

yet below EDIFFG after NSW ionic steps, the relaxation continues from its last structure (CONTCAR), up to three times.

poscar0 The relaxed cell is reduced and oriented again and written as `POSCAR0`: every later step uses this cell and this Cartesian frame. The report gives its space group and lattice parameters.

scf A self-consistent calculation on POSCAR0 in `02_scf` (same INCAR, no ionic steps, same k-mesh), which also gives the band gap. If the cell did not change in the previous task, it starts from the charge density and wave functions of the relaxation.

dfpt For an insulator (band gap above 0.05 eV, `DFPT: auto`): density-functional perturbation theory in `02_scf/dfpt` gives the high-frequency dielectric constant and the Born effective charges of every atom (Section 8). `DFPT: on` forces it, `off` skips it.

snaps The supercell is chosen: the multiple of the conventional cell with the fewest atoms whose faces are at least 10 Å apart (`SUPERCELL: SMALL`, at most 100 atoms; `LARGE`: 14 Å, at most 300 atoms); if none is that large, the one with the largest distance. SC-SNAPS (`03_snaps`) then writes the undistorted supercell `poscar_000` and 50 snapshots, each atom displaced as in a thermal ensemble of harmonic oscillators of frequency `AVG_FREQ` at temperature `TEMPERATURE`. The temperature is adjusted (SC-SNAPS rerun at most twice) until the rms displacement is within 15 % of `DISP_TARGET` (0.02 Å): large enough for clean forces, small enough for the Taylor expansion of the energy. The undistorted supercell is checked to be an exact multiple of POSCAR0 before any VASP job is run.

supercells One folder per snapshot, `04_supercells/snap_NNN`, with its POSCAR, the SCF INCAR without symmetry (`ISYM = 0`) and a k-mesh equivalent to the SCF one (Section 4.3); magnetic moments given per atom are carried over to the supercell.

vasp_snaps The VASP runs of the snapshots, one after the other. A run that fails does not stop the others; the failed ones are listed in `report.log`, and `resubmit_failed.sh` reruns only those. An existing OUTCAR is never overwritten (it is kept as `OUTCAR.1`, ...).

collect `read_outcar.x` reads the positions and forces of every run into `04_supercells/FORCEDISP1`; `POSCAR1` is the undistorted supercell. The report gives the largest net force on a supercell (should be ≈ 0) and the largest difference between the positions in OUTCAR and in POSCAR.

focex The five input files of FOCEX are written in `05_focex` (`structure.params`, `latdyn.params`, `kpbs.params`, `dielectric.params`, `default.params`) and FOCEX is run. With Born charges a second run without them is made in `05_focex/NOBORN`, for comparison; with fewer than `LASSO_RATIO` (2) equations per unknown, a third run with LASSO, which keeps only the force constants the data need, is made in `05_focex/LASSO`. The report gives the fit error, the numbers of equations and unknowns, the lowest ω^2 (a warning below $-25\,\mathrm{cm}^{-2}$: imaginary modes) and the highest frequency. If the DOS range of `latdyn.params` does not fit the frequencies, FOCEX is run again with one that does.

plots The gnuplot scripts `bvel.plt`, `all.plt`, `qha.plt` are filled in and run in every FOCEX folder (Section 9); `makeplots.sh` remakes them.

4.1 Defaults

- `ACCURACY: Accurate`: ENCUT 600 eV, EDIFF 10^{-7} eV, EDIFFG -0.001 eV/Å, PREC Accurate in every VASP step. `Normal`: 450 eV, 10^{-6}, -0.003, PREC Normal – cheaper,

but the residual forces of the relaxed structure are larger (0.01 eV/Å in a ZnO test), which adds noise to the force constants. DFPT uses EDIFF 10^{-8}.

- Relaxation and SCF k-mesh: VASP-Flow's `coarse` density (1000 k-points per reciprocal atom), `fine` (6000) if every element is a metal.
- 50 snapshots. Each gives three equations per atom of the supercell: 20 snapshots of a 96-atom supercell give 5760 equations for the 264 unknowns of ZnO with ranks 1–3, so fewer snapshots are often enough (`SNAPSHOTS`).
- `RANGE: 6 2 1 0 0 0 0`: neighbour shells kept for ranks 2, 3, 4, ...; `FC2RANGE: 0`: the harmonic force constants are kept over the whole supercell.
- FOCEX k-mesh `fine`; the band-structure path is found automatically.

4.2 The FOCEX frame

FOCEX and SC-SNAPS build the conventional cell from $a, b, c, \alpha, \beta, \gamma$ with a along x and b in the xy plane, and the supercells, forces and Born charges must be in that same Cartesian frame. The agent rotates the cell into it at the start, so everything VASP computes is already in the right frame. `structure.params` lists the conventional cell (line 1), the primitive vectors in its units (line 2) and the atoms in conventional coordinates.

4.3 The k-mesh of the supercells

If the supercell is a diagonal multiple of the primitive cell, its mesh is the SCF mesh divided by the multiples, rounded up (6 6 4 with a $3 \times 3 \times 2$ supercell gives 2 2 2). Otherwise (every multiple of a centred conventional cell) it keeps the k-point spacing of the SCF mesh. `KPOINTS_snaps` sets it yourself (Section 7).

5 Mode extract_only: FOCEX on existing data

Use it when the supercell forces exist already – from an earlier calculation, another code, or a colleague – and you want FOCEX's force constants and phonons. No VASP and no SC-SNAPS are run.

The data folder (named in the instruction file with `DATA:`) holds

- `POSCAR1` and `FORCEDISP1`: a supercell and the positions and forces of its snapshots, in FOCEX's format (what `read_outcar.x` or `read_qe.x` write); more pairs `POSCAR2`/`FORCEDISP2`, ... are all used. Any orientation.
- optionally `POSCAR`: the primitive cell, in the same orientation as the data; without it, spglib finds the primitive cell of `POSCAR1`.
- optionally `dielectric.params`: the dielectric constant and the Born charges, in the frame of the data, one 3×3 block per atom of that primitive cell. With more than one atom of a species the primitive `POSCAR` is required, to fix the order of the atoms.

`SPECIES: Mg O` names the elements when the POSCARs have no element line.

What the agent does. At set-up it symmetrizes the lattice of the primitive cell (DFT lattice parameters given to 6–7 digits make FOCEX's symmetry analysis lose force constants), finds the rotation into the FOCEX frame, and rotates the supercells, the positions and forces in `FORCEDISPi`, the dielectric constant and the Born charges with it. The report gives the rotation and the size of the symmetrization. The workflow then has only the tasks focex and plots.

```
cat > focex-agent-instructions.txt <<'EOF'
Project: MgO
MODE: extract_only
DATA: ~/data/MgO_forces          # POSCAR1, FORCEDISP1, dielectric.params
SPECIES: Mg O
RANKS: 1 1 1 0 0 0 0 0
EOF
python ~/Anharmonic-lattice-dynamics/FOCEX/AGENT/focex-agent.py
cd MgO && bash run_workflow.sh          # FOCEX takes minutes: no batch job needed
```

Checked on MgO data given in a rhombohedral orientation: the fit error (7.197 %), the 35 unknowns, the frequencies and the elastic moduli are those of a FOCEX run in the original frame.

6 The instruction file: every keyword

One keyword per line, `KEYWORD: value`; `##` starts a comment. The VASP keywords are those of VASP-Flow (its manual lists all of them); the common ones are below.

keyword	default	meaning
project		
`Project`	(required)	a name, used in titles and as the default folder
`PROJECT_DIR`	the Project name	where the calculations go; relative to the instruction file
`MODE`	from_scratch	or extract_only
`DATA`	–	extract_only: the data folder
`SPECIES`	–	extract_only: element names, if the POSCARs have none
`PROFILE`	none	the VASP-Flow profile, as `--profile`
force constants		
`RANKS`	(required)	8 flags for ranks 1–8: `1 1 0 0 0 0 0 0` harmonic, `1 1 1 0 0 0 0 0` + cubic, `1 1 1 1 0 0 0 0` + quartic
`RANGE`	6 2 1 0 0 0 0	neighbour shells for ranks 2–8
`FC2RANGE`	0	0: harmonic range = the supercell
`LASSO_RATIO`	2	below this many equations per unknown, an extra LASSO run
VASP (VASP-Flow keywords)		
`FUNCTIONAL`	R2SCAN	PBE, PBEsol, LDA, AM05, R2SCAN, HSE06, VV10
`MPI`	the profile's	MPI ranks for VASP
`ACCURACY`	Accurate	or Normal (Section 4)
`ENCUT`	by ACCURACY	plane-wave cutoff (eV)
`KMESH_DENSITY`	coarse or fine	k-mesh of the relaxation and SCF
`MAGMOM`	–	magnetic moments, e.g. `MAGMOM: Cr=3.0`

keyword	default	meaning
GGA_U	auto	Hubbard U from VASP-Flow's table: auto applies it to d/f elements when the compound has O, S, Se, Te or a halogen; ON always; OFF never
DFPT	auto	Born charges: auto (if there is a gap), on, off
BORN_TRANSPOSE	yes	Section 8
snapshots		
SUPERCELL	SMALL	SMALL, LARGE, or 9 integers (supercell vectors as rows, primitive units)
SNAPSHOTS	50	number of snapshots
TEMPERATURE	30	K (adjusted to DISP_TARGET)
AVG_FREQ	400	cm^{-1}, the frequency of the oscillators of SC-SNAPS
DISP_TARGET	0.02	Å, the rms displacement wanted
FOCEX		
FOCEX_MESH	fine	k-mesh of latdyn.params: coarse, fine, a density, or 3 integers
KPATH	0	0: automatic band path; 1: give a kpbs_params block
BS_NKPTS	40	points per segment of the band path
programs		
FOCEX_X	newest ~/ALADYN/bin/ focex-*.x	the FOCEX binary
SC_SNAPS_X	~/BIN/sc_snaps.x	SC-SNAPS
READ_OUTCAR_X	~/ALADYN/bin/ read_outcar.x	reads OUTCARs

7 Changing the files the agent writes

Every file the agent generates can be changed *from the instruction file*, with a block named after the file: a line with its name, the contents, and a line with the name followed by _END.

- **INCAR blocks are merged tag by tag**: the tags you give replace or add to those the agent writes; the others stay. INCAR applies to every VASP step; INCAR_relax, INCAR_scf, INCAR_dfpt, INCAR_snaps to one.
- **Every other block replaces the whole file**, so it must be complete: KPOINTS (relaxation and SCF), KPOINTS_snaps, cell_inp, supercell_inp, snaps_inp (SC-SNAPS), structure_params, latdyn_params, kpbs_params, dielectric_params, default_params (FOCEX).

Simple choices – the supercell, the number of snapshots, the FOCEX k-mesh – have their own keyword; use a block only for what has none.

Example. Wurtzite ZnO without Hubbard U (VASP-Flow would otherwise put its tabulated U on Zn, an oxide), LASPH in every VASP step, stricter electronic convergence only in the supercell runs, a $3 \times 3 \times 2$ supercell, 20 snapshots, a denser FOCEX mesh, and your own band path Γ–M–K–Γ–A, which replaces kpbs.params:

```
Project: ZnO
RANKS: 1 1 1 0 0 0 0 0
FUNCTIONAL: PBEsol
GGA_U: OFF
MPI: 40
SUPERCELL: 3 0 0  0 3 0  0 0 2       # rows: supercell vectors in primitive units
SNAPSHOTS: 20
FOCEX_MESH: 12 12 8
KPATH: 1                              # the band path is given below

INCAR                                 # every VASP step (merged tag by tag)
  LASPH = .TRUE.
INCAR_END
INCAR_snaps                           # the supercell runs only
  NELM = 200
  ALGO = All
INCAR_snaps_END
kpbs_params                           # replaces kpbs.params (FOCEX's band path)
1        # coordinates in units of the primitive reciprocal vectors
60       # k-points per segment
4        # number of segments
G   0.000000 0.000000 0.000000
M   0.500000 0.000000 0.000000
K   0.333333 0.333333 0.000000
G   0.000000 0.000000 0.000000
A   0.000000 0.000000 0.500000
kpbs_params_END
```

After the set-up you can read `01_relax/INCAR` before anything runs.

8 Born charges and the dielectric constant

In an insulator, the long-range Coulomb interaction between the displaced ions splits the longitudinal and transverse optical modes near Γ (LO–TO splitting). FOCEX includes it with the Born effective charges Z^* and the high-frequency dielectric constant ε_∞; without them the optical branches near Γ are wrong.

- from_scratch: the agent computes them (task dfpt) when the SCF shows a band gap, writes them to `dielectric.params`, and makes for comparison a FOCEX run without them in `05_focex/NOBORN`. `DFPT: off` skips them; `on` computes them for any material.
- extract_only: give them in `dielectric.params` in the data folder.
- your own values, in either mode: a `dielectric_params` block (it replaces the file).

Note. VASP prints Z^*_{ij} with i the direction of the field and j that of the displacement; FOCEX expects the transpose, which the agent writes. `BORN_TRANSPOSE: no` writes them as VASP prints them; you should not need it.

9 The project folder and the results

file or folder	what it is
`focex-agent-instructions.txt`	the copy of your instruction file that the runs read

file or folder	what it is
report.log	one entry per task: OK or FAILED, minutes, what was checked – read this first
state.json	the finished tasks and the numbers passed between them
run_workflow.sh	the job; resubmit_failed.sh reruns the failed supercell runs
POSCAR.input, POSCAR.init, POSCAR0	your structure; reduced and oriented; relaxed
01_relax, 02_scf, 02_scf/dfpt	VASP: relaxation, SCF, Born charges
03_snaps	SC-SNAPS input and the snapshots poscar_NNN
04_supercells	snap_NNN/ (one VASP run each), POSCAR1, FORCEDISP1
05_focex	FOCEX: inputs, outputs, plots; NOBORN/ and LASSO/ when made
diagnosis	written by diagnose.py: report.txt, fix_it.sh, backups

In 05_focex: the force constants fc2.dat, fc3.dat, ... with the cell in lat_fc.dat; the dispersion bs_freq.dat; the elastic constants and moduli mech.dat; thermodynamics and thermal expansion thermo_QHA.dat; and three PDFs: <Project>.pdf (the dispersion coloured by group velocity, and the group velocities against frequency), all-<Project>.pdf (Grüneisen parameters, density of states, dispersion) and qha.pdf (thermal expansion, Grüneisen parameter, bulk modulus, thermodynamic functions, free energy against volume). The FOCEX manual describes every output file. The large VASP files (WAVECAR, CHGCAR, CHG, ...) are deleted when no longer needed; the OUTCARs never.

10 When a run stops: diagnose.py

```
A=~/Anharmonic-lattice-dynamics/FOCEX/AGENT
python $A/diagnose.py ~/work/ZnO/ZnO     # one or more project folders
less ~/work/ZnO/ZnO/diagnosis/report.txt
bash ~/work/ZnO/ZnO/diagnosis/fix_it.sh
```

diagnose.py itself changes nothing. It lists every task as done, FAILED, INTERRUPTED (the job was killed inside it) or not reached, or reports that the project is still running or not yet started. For a failed task it finds the cause from report.log, the OUTCARs, vasp.out, focex.out and the SLURM logs, and writes diagnosis/report.txt (what failed, where, the evidence, what the fix will change) and diagnosis/fix_it.sh. VASP failures are recognised with the rules of VASP-Flow's diagnosis tool; FOCEX, SC-SNAPS and read_outcar.x stops by their messages.

cause	meaning	what fix_it.sh does
`timeout`	the job hit its time limit	walltime ×2, resubmit
`scf_nelm`	SCF not converged in NELM steps	ALGO = All, NELM = 200
`edddav`	the Davidson eigensolver failed	ALGO = All
`zbrent`	the relaxation's line search failed	IBRION = 1
`relax_unconv`	relaxation not converged	resubmit (continues from CONTCAR)
`oom`	out of memory	KPAR = 1
`mpi_init`, `not_run`, `interrupted`	MPI did not start; never ran; killed	resubmit
`ws_outside`	FOCEX `R+tp-t not inside!`: the supercell lacks the symmetry of the lattice	held: choose another supercell, redo from the snapshots
`others`	disk quota, symmetry, an unknown stop, ...	held, with the error lines

New INCAR tags go into the INCARs of the failed runs and into the matching block of the project's instruction file (`INCAR_snaps`, ...), so that later runs use them too. Before it changes anything, `fix_it.sh` checks that nothing is running and that nothing changed since the diagnosis, and copies every file it edits to `diagnosis/backup_<date>/`; then it resubmits (`sbatch` with a SLURM profile, otherwise `bash` in the background). When the fix needs your judgement – a cause it does not know, a fix that was already tried, a new supercell – the script is *held*: it prints what to do and stops, and `bash fix_it.sh -anyway` runs its suggested commands as they are. Run `diagnose.py` again when the resubmitted job has finished.

Example. ZnO with a $4 \times 3 \times 2$ supercell: *task focex, 05_focex: ws_outside – an FC2 pair of a kept symmetry group lies outside the Wigner-Seitz cell of the supercell ...; evidence: focex.out: "SET_DYNMAT_STD: R+tp-t not inside!"; HELD for review: use a supercell with the symmetry of the lattice, then redo from the snapshots.*

11 Thermal conductivity

THERMACOND reads `lat_fc.dat`, `fc2.dat` and `fc3.dat` from `05_focex`, so fit at least the cubic force constants (`RANKS: 1 1 1 0 0 0 0 0`). The agent does not run it: in `05_focex` set lines 8–13 of `latdyn.params` (see the THERMACOND manual) and run `tcond-2.0.x` there.